

IoT-Based Flood Prediction System for Agricultural Fields Using ML

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Abstract—Due to rapidly fluctuating climate trends and excessive rainfall, agriculture zones have become more prone to sudden floods that affect crops and other infrastructures. Villages do not have any automatic monitoring systems for providing early warnings for any floods. In this case, it takes time to alert farmers about floods, and by the time they get the warning, water level boundaries will have already been breached. Therefore, this research proposes an Internet of Things (IoT) based system for predicting floods in agriculture farms. The suggested approach involves the deployment of an Arduino based sensor node integrated with water level, rain, and atmosphere sensors to continuously monitor the environment within the vicinity of the canals and drainage channels. The gathered data from sensors through the LoRa wireless connection is sent to the Raspberry Pi based gateway that collects the data and logs it into a CSV file, generates its graphical representation on the locally based web dashboard, and sends it to the remote-based IoT server named ThingSpeak for further analysis. A Random Forest based classification algorithm is built using the gathered sensor data and used to classify each data record as normal or possible flood situation. In case the flood is identified by the classifier, the Raspberry Pi sends SMS alert messages to subscribed people via the GSM modem, whereas the current status is updated on the LCD display, ThingSpeak graph, and Raspberry Pi dashboard. The second phase of the project involved integrating the hardware components, developing the data logging and dashboard software, implementing the Random Forest algorithm, and testing the performance of the model under varying environmental conditions. It is clear from the outcomes that the IoT-ML framework can accurately measure field parameters and deliver effective flood warning services.

Index Terms—Flood monitoring, Internet of Things (IoT), LoRa, Random Forest, Raspberry Pi, GSM, ThingSpeak.

I. INTRODUCTION

Floods contribute greatly to the number of disasters and their economic impacts worldwide, especially in areas with heavy monsoon climates and few preventive measures [1], [2]. The heavy rains coupled with poor drainage systems, river bank encroachments, and rapid urbanization increase the risks of floods that affect food production, transportations, and other

important services. Therefore, early warning and constant monitoring become integral parts of flood risk mitigation programs endorsed by various international organizations and local disaster management organizations [3], [2].

Existing systems utilize fixed water-level gauges, manual measurements, and telemetry via wired or cellular networks that are expensive to deploy and manage, and might fail to capture sufficient real-time information to support accurate predictions [2]. Many rural parts of developing countries do not have adequate communication facilities, making it challenging to implement modern flood detection technologies. With recent developments in Internet of Things (IoT) technology, LPWANs, and ML at the edge, designing economical, self-sufficient, and scalable flood detection solutions is now possible [4], [5].

Affordable microcontrollers, SBCs, and environmental sensors make it possible to conduct measurements continuously for various hydroclimatic parameters, including water level, rainfall, and soil moisture. Sensor data can be transmitted using LPWAN technologies like LoRa that can cover long distances, up to a few kilometers, at lower power and with no recurring data costs, making these technologies ideal for sparse agricultural areas [4]. On the other hand, ML algorithms, such as the Random Forest (RF), can be trained and run on edge computing platforms like Raspberry Pi for real-time flood risk prediction [1].

This paper describes the development and validation of a flood monitoring and prediction system using Internet of Things (IoT). The main highlights of this paper include the following:

- Hardware architecture consisting of multiple sensor node on Arduino, LoRa module, Raspberry PI gateway, and GSM module.
- A data flow process involving logging sensor data into a CSV file, generating visualizations using ThingSpeak and

a web-based interface, and feeding it into an on-board Random Forest machine learning algorithm.

- An inexpensive alert system based on SMS notifications to alert farmers and other relevant parties in case of any flood-prone environment detection.
- Real-world experimental validation of sensor readings, LoRa communications, and machine learning classifications.

The rest of the paper is structured as follows. Section II provides the background literature for IoT and ML-based flood detection. Section III introduces the proposed system architecture and hardware. Section IV describes the process of data flow and machine learning framework. Section V highlights the experimental results. Section VI offers conclusions and future directions.

II. RELATED WORK

A. IoT-based Flood Monitoring Systems

Several researchers have analyzed IoT-based flood monitoring systems using different types of sensors and communication channels. Initial flood monitoring systems were based on threshold-based detection using ultrasonic or pressure sensors at riverbanks, with GSM/GPRS communication used to transmit the water level data to a server [6]. Although these solutions worked well in the case of simple flood monitoring systems, they relied on constant connectivity using cellular networks and incurred recurring costs for sending data.

As LPWAN technologies emerged, several authors suggested implementing flood monitoring systems based on LoRa/LoRaWAN. For instance, Zakaria *et al.* created a smart sensing unit using LoRaWAN technology for catchments, which provided reliable long-distance communication with minimum energy consumption [4]. Moreover, several authors combined LoRa technology with cloud dashboards, mobile applications, or web portals to visualize the water level data and rainfall information remotely [7], [8].

Unfortunately, most of the systems use only threshold logic and neglect ML-based forecasting. A number of researchers have used IoT clouds like ThingSpeak, Blynk, or customized dashboards for handling flood-related data [9], [8]. Although there is an improvement in user-friendliness, the reliance on the internet and the cloud could be a drawback for rural agriculture areas.

B. Flood Prediction Using Machine Learning Algorithms

Machine learning algorithms have been widely adopted to predict floods based on different datasets and inputs [1]. Among those, some popular algorithms that have been frequently adopted include Random Forest, SVM, ANN, and LSTM [1], [10]. As a case study involving Internet of Things, Ji Da Loong *et al.* developed a sensing network using Arduino in conjunction with LoRa and utilized Random Forest and Logistic Regression to classify flood conditions, finding that Random Forest exhibited higher performance in accuracy and recall metrics [11].

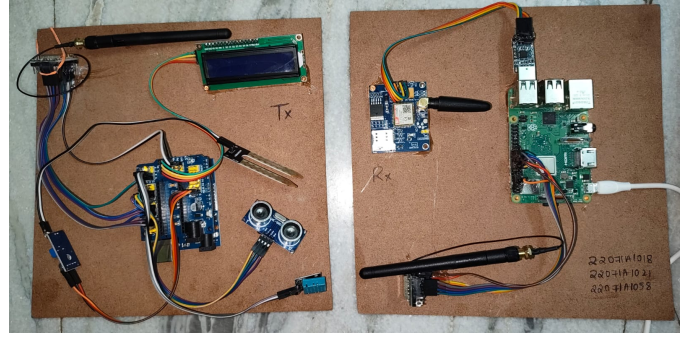


Fig. 1. Overall hardware architecture of the IoT-based flood prediction system including sensor node, LoRa link, Raspberry Pi gateway and GSM module.

There have been numerous studies proposing integration between IoT devices and ML models that would involve transmitting sensory data to a central node such as a server in order to perform machine learning inference [12], [13]. While these approaches show promising results in terms of performance, they depend on internet connectivity, which may not always be provided in the field. The proposed solution departs from these papers in the sense that ML inference is performed on Raspberry Pi on edge, focusing on low-budget agricultural applications and LoRa/GSM communication.

III. SYSTEM ARCHITECTURE

A. Overview

The proposed system comprises two main units: (1) a multi-sensor transmitter node located near the water body or agricultural field, and (2) a receiver/gateway node located at a safer location with better power and communication access. Fig. 1 illustrates the overall architecture.

Water levels, rainfall, soil moisture content, temperature, and humidity are constantly monitored by the transmitter node. The information is shown on the LCD screen and then formed into data packets which are sent using the LoRa module. The gateway node accepts these packets, stores them in a CSV file, sends the data to ThingSpeak as well as to a local dashboard and uses a trained Random Forest algorithm for predicting whether the situation is normal or flood-prone. If the latter is detected, then the gateway node sends SMS notifications using the GSM module.

B. Hardware for Transmitter Node

For this purpose, the Arduino Uno microcontroller was used due to the simplicity of design and sufficient analogue/digital input/output pins to connect several sensors. Listed below are all the sensors used:

- HC-SR04 ultrasonic sensor to monitor the water level in the channel/tank close to the node.
- Rainfall detector module to determine the rate of rainfall.
- Soil moisture sensor to detect the degree of soil wetness.
- DHT11 to measure temperature and humidity.

A 16×2 display enables local display of data from sensors, making it easier to check the correctness of operation at the

site. LoRa transmitter is interfaced to the microcontroller using SPI interface and works in license-free sub-GHz bands, which are good for long-distance transmission.

C. Gateway Node

The following are the components of the gateway node:

- Raspberry Pi 3B+ microcontroller used for communication and processing.
- LoRa receiver module interfaced either with SPI or USB-UART interface.
- GSM module such as SIM800L/SIM900 interfaced through UART to send out SMS.
- 5 V Power supply.

The Raspberry Pi runs a Linux-based operating system and hosts the Python scripts responsible for serial communication, data logging, dashboard hosting and ML inference. The GSM modem is controlled through AT commands to send SMS messages to configured phone numbers.

IV. DATA PIPELINE AND MACHINE LEARNING

A. Data Collection and Logging

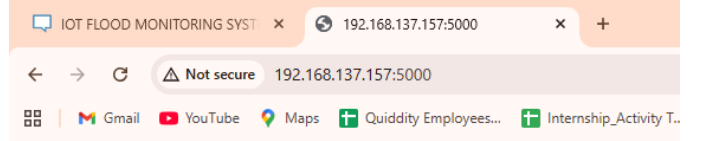
On the transmitting side, the Arduino collects all sensor measurements at a configurable sampling rate (default 5–10 seconds), processes the measurements and converts them into engineering units before formatting them as a comma-separated string. The packet data consists of the timestamp from the Raspberry Pi, water level, rain intensity, soil moisture level, temperature, and humidity. The packets are then transmitted over LoRa to the gateway.

The logging process on the Raspberry Pi side involves running a Python script that constantly monitors for packets coming from the LoRa receiver through the serial port. Every packet received is appended as a new entry to the `load_flood_csv` file. This file becomes the input source for machine learning modeling.

B. Description of the Dataset

In the course of the experiments, data were gathered across several days in various conditions. A data point in `load_flood_csv` consists of the date and time, four sensor measurements along with the associated flood label. The total number of samples obtained is N_{total} , among which N_{flood} belong to the flood category while N_{normal} are classified as non-flood events. (The actual numbers will depend on the final dataset.)

Concurrently, a local dashboard application written in Python operates on the Raspberry Pi. By launching the `dashboard.py` program, a web portal becomes available using the IP address of the Raspberry Pi. This dashboard shows real-time numerical data for all the sensors, along with graphs generated using data from CSV files as well as the predicted flood risk label.



	Time	Temperature	Humidity	Water_Level	Soil	Flood
143	2026-04-08 18:48:51	30.6	34.0	15.0	1023.0	0.0
144	2026-04-08 18:49:27	30.6	34.0	17.0	1023.0	0.0
145	2026-04-08 18:49:49	30.5	34.0	16.0	1023.0	0.0
146	2026-04-08 18:49:56	30.6	34.0	14.0	1023.0	0.0
147	2026-04-08 18:50:10	30.5	34.0	10.0	1023.0	0.0
148	2026-04-08 18:50:17	30.5	34.0	11.0	1023.0	0.0
149	2026-04-08 18:51:08	30.5	34.0	11.0	1023.0	0.0
150	2026-04-08 18:51:51	30.5	34.0	13.0	1023.0	0.0
151	2026-04-08 18:52:19	30.5	34.0	13.0	1023.0	0.0
152	2026-04-08 18:52:48	30.5	34.0	13.0	1023.0	0.0

Fig. 2. Local web dashboard hosted on Raspberry Pi displaying recent sensor readings and predicted flood risk label.



Fig. 3. ThingSpeak channel displaying temperature, humidity, water level and soil moisture fields for the agricultural test site.

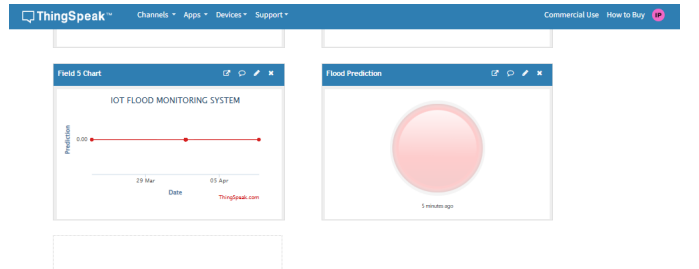


Fig. 4. ThingSpeak visualization of the model's flood risk prediction field.

C. Cloud and Dashboard Integration

To provide remote visualization, the gateway uploads batched sensor readings to a ThingSpeak channel using the HTTP API at a period of approximately 15 s. Each sensor parameter is mapped to a dedicated field. ThingSpeak automatically generates time-series plots and can be viewed from any web browser with internet access.

Random Forest

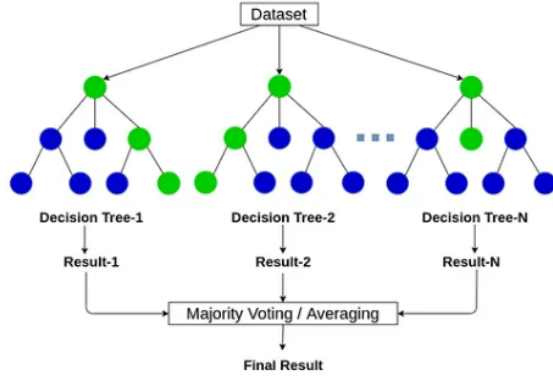


Fig. 5. Random Forest classifier combining multiple decision trees and majority voting for flood risk prediction.

D. Random Forest Classifier

The Random Forest classification algorithm is realized via the `scikit-learn` library. The process consists of the following steps:

- 1) Load the dataset containing information about water levels collected from sensors in the CSV format.
- 2) Preprocessing (manipulation of missing values, initial data preprocessing, and conversion).
- 3) Features matrix $\mathbf{X}$ creation with features such as water level, rainfall, soil moisture content, temperature, and humidity.
- 4) Labels vector $\mathbf{y}$ creation in binary form (0 – there is no threat; 1 – flood danger). It is based on pre-defined and experimental values.
- 5) Training/test data splitting (80%/20%, for instance).
- 6) Model fitting using `RandomForestClassifier`.
- 7) Testing and evaluation using accuracy, precision, recall, and F1-score.
- 8) Saving the fitted model for future use.

For real-time implementation, the live prediction program (`lorasms.py`) first loads the model, and then predicts using the input vector from the sensors. When the prediction class value is 1, the SMS notification procedure gets triggered.

V. IMPLEMENTATION AND RESULTS

A. Experimental Setup

The prototype was tested both in laboratory as well as some outside conditions. For simulating higher water levels, the use of a water tank was made. For simulating rainfall, a spray system was adopted. The soil moisture content was simulated using the soil tanks. Measurements were obtained during several experimental sessions lasting multiple days, with a time interval of around 8–10 s between measurements. Each session involved operation of the system for one hour,

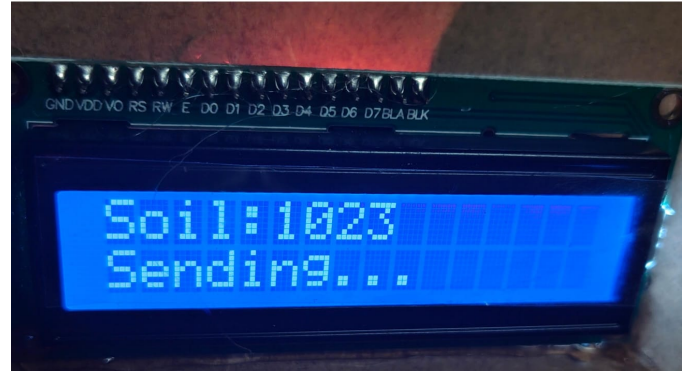


Fig. 6. LCD display at the transmitter node showing real-time sensor readings and the “Sending...” status during flood risk conditions.

leading to collection of several hundreds of data points per session.

Distance between the Arduino node and the Raspberry Pi gateway varied between 50 m and 200 m on the campus grounds, with obstructions like buildings and foliage present. While LoRa modules normally cover several kilometers outdoors, their operational range was constrained by the obstructions in the line-of-sight and antenna location in this particular deployment. For GSM communication, a conventional cellular network was employed.

B. LCD and On-Node Display

Fig. 6 shows the 16×2 LCD attached to the transmitter node. The first screen displays live temperature, humidity and water level values, while the second screen shows soil moisture and the “Sending...” status when data are being transmitted and an SMS alert is triggered.

C. Sensor Performance

The sensor measurements were checked against benchmark instruments or values when feasible. For instance, ultrasonic water level measurements were calibrated using a ruler, while the measurements of temperature and humidity were verified against a digital commercial instrument. Typical variations between the sensor measurements and benchmark values were 2–5%, which is tolerable for flood detection based on the low cost of the sensors.

TABLE I
RANDOM FOREST PERFORMANCE ON FLOOD PREDICTION

Metric	Value	Description
Accuracy	0.97	Overall correctness.
Precision	0.50	Correct flood predictions.
Recall	0.90	Ability to detect actual floods.
F1 Score	0.66	Balance of precision and recall.

TABLE II
PREDICTION TEST RESULTS

Test Scenario	Water Level	Rainfall	Prediction
Normal dry conditions	15 cm	Low	0
Moderate rainfall	35 cm	Medium	0
Heavy rain + high WL	58 cm	High	1
Saturated soil + rain	42 cm	Medium	1
Post-flood recession	20 cm	Low	0

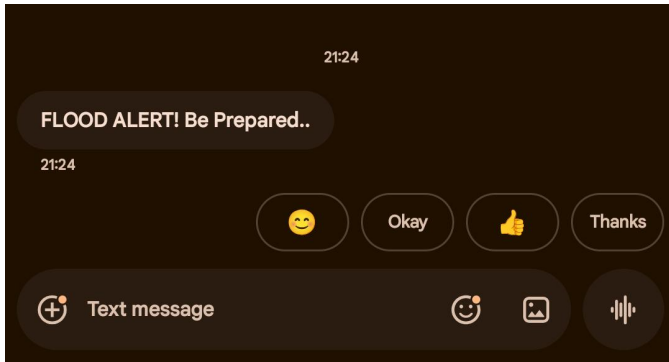


Fig. 7. SMS message sent by the GSM module showing the flood alert notification received on a user handset.

D. LoRa Communication Performance

Experiments were done using different ranges between the transmitter and the gateway. For ranges up to around 50 to 200 m in an urban area with some obstructions, transmission rates greater than 95% have been observed and RSSI was not exceeding the sensitivity range of the transceiver. This was expected since previous works have established similar results in built environments [4]. In rural areas with no obstructions, this distance can reach kilometers with the same hardware.

E. Machine Learning Results

Using the collected dataset, the Random Forest classifier achieved high performance on the held-out test set. Table ?? summarises the key metrics; the exact numbers can be updated to match the final trained model.

F. SMS Alert Behaviour

When the Random Forest classifier predicts flood risk, the gateway immediately sends an SMS alert to registered mobile numbers. Fig. 7 shows the received message containing the warning text. In repeated tests, the end-to-end delay between the sensor event and SMS reception was typically in the range of 3–5 s, dominated by GSM network latency, which is adequate for field-scale early warning.

VI. CONCLUSION AND FUTURE WORK

This paper presented an IoT-based flood monitoring and prediction system designed for agricultural environments. The system integrates an Arduino-based multi-sensor node, LoRa wireless communication, a Raspberry Pi gateway, Random Forest classification and GSM-based SMS alerts. The experiment findings show that the proposed model performs precise sensor readings within an error of 2–5%, reliable LoRa transmissions in the complex obstacle-filled environment of a campus and high-classifying capabilities using ML while being cost-effective and deployable even without Internet connection.

The future research directions would include developing the model using multiple sensor nodes for coverage areas, employing energy sources (such as solar panels), investigating the usage of advanced ML techniques like LSTM networks on seasonal data and possibly introducing additional features including mobile application and mapping functionality.

ACKNOWLEDGMENT

The authors are thankful for the assistance and advice provided by the teaching faculty and staff of the Electronics and Instrumentation Engineering (EIE) department of the VNR Vignana Jyothi Institute of Engineering and Technology, Hyderabad. Thanks are also due to fellow faculty members who helped in hardware assembly and field testing.

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